

Code File Index

[Complete sequence of all 19 code files from V1 to final](#)

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Overview

This repository contains 19 Python scripts organised in chronological experiment order. The files are divided into Phase 2 (feature engineering and data splitting) and Phase 3/3+4 (contrastive autoencoder training, detection, and explanation). Each version addresses one specific hypothesis or fixes one identified problem.

Phase 2 Files — Feature Engineering and Splitting

Phase 2 loads the CSE-CIC-IDS2018 dataset, applies CADE's feature engineering (83 features), and creates the train/test split. Phase 3 reads the output (`working_set.parquet`). Run Phase 2 first.

File	V	Split method	Features	Key change	Used with
phase2_v1.py	V1	Global temporal	82	Global sort all 147k, cut at 80% — most attack classes land entirely in train	phase3_v1/v2
phase2_v2.py	V2	Global temporal	82	Same as V1 — minimal changes for Phase 3 compatibility	phase3_v2
phase2_v3.py	V3	Global temporal	82	Same split, tested with <code>BALANCED=False</code> in Phase 3	phase3_v3
phase2_v4.py	V4	Global temporal	82	Same split, tested with L2-norm encoder in Phase 3	phase3_v4
phase2_v5.py	V5	Global temporal	82	Same split, tested wrong <code>similar_ratio</code> in Phase 3	phase3_v5
phase2_v6.py	V6	Global temporal	82	Same split, tested correct <code>similar_ratio</code> in Phase 3	phase3_v6
phase2_v7.py	V7	Random stratified	82	<code>sklearn train_test_split</code> — equal class% in both splits	phase3_v7
phase2_v10.py	V10	Within-class temporal	83	Each class sorted independently. First 80% train, last 20% test. Adds <code>Timestamp_ms</code> (83rd feature). Used with 0/100 hold-out in Phase 3.	phase3_v10, phase34_v11, phase34_v12

Phase 3 Files — Detection Only (V1–V10)

Phase 3 trains the contrastive autoencoder and runs IQR-based drift detection. These files implement only detection (no explanation). Phase 3+4 files (below) add Algorithm 2 explanation and T-SNE visualisation.

File	V	Distance	Loss	Balanced	ratio	lr	Key change
phase3_v1.py	V1	d ²	sum	True	all	0.001	Baseline: squared distance, all-pairs sum, 80/20 hold-out
phase3_v2.py	V2	d ²	sum	True	all	0.001	Same as V1 — Timestamp handling update only
phase3_v3.py	V3	d ²	sum	False	all	0.001	BALANCED=False — worse results, Infiltration F1 dropped to 0.118
phase3_v4.py	V4	d ²	sum	True	all	0.001	L2 norm on encoder output — BROKEN, all stuck contras≈4.67
phase3_v5.py	V5	d ²	mean	True	0.25*	0.001	*Wrong ratio: n_neg_target > n_neg so all negatives used. Ratio stays 33%
phase3_v6.py	V6	d ²	mean	True	0.25	0.001	Correct ratio: subsample positive pairs. n_pos=7,225 from 10,710
phase3_v7.py	V7	d ²	mean	True	0.25	0.001	Same as V6, used with V7 random-split parquet
phase3_v8.py	V8	d ²	mean	True	0.25	0.0001	lr=0.0001 (CADE's value) — worse, insufficient convergence in 300 epochs
phase3_v10.py	V10	d	mean	True	0.25	0.001	Regular L2 + 0/100 hold-out. BEST standalone detection: Avg F1=0.842

Phase 3+4 Files — Detection + Explanation + T-SNE

These scripts combine detection (Phase 3), explanation (Algorithm 2, Phase 4), and T-SNE visualisation in a single run. Use these for the complete pipeline.

File	V	Distance	ratio	m	Key change	Avg F1	Infil crossing
phase34_v11.py	V11	d	all-pairs	10	Exact paper spec: all-pairs mean, no subsampling, regular L2, 0/100 hold-out	0.849	14%
phase34_v12.py	V12	d	0.10	10	Reduces positive pairs to 10% — targets wider IQR. Bot/DoS-Hulk crossing excellent	0.851	3%

How to Run (Recommended Sequence)

To reproduce V10 — best combined results (Avg F1=0.842, crossing=40.5%):

1. Run **phase2_v10.py** → produces working_set.parquet (83 features, within-class temporal split)
2. Run **phase3_v10.py** → detection only. Avg F1=0.842, Infiltration crossing=40.5%

To reproduce V11 — exact paper specification:

1. Run **phase2_v10.py** (same parquet)
2. Run **phase34_v11.py** → detection + explanation + T-SNE. Avg F1=0.849

To trace the full experimental journey from V1:

Run phase2_v1.py → phase3_v1.py, then progress through versions sequentially. Each version's docstring explains what changed and why.